

Marine Explanation Accuracy and Transfer

A. Evidence-calibrated explanations across land and marine domains

The marine evaluation tests whether the paper’s central explanation behaviors transfer to autonomous surface navigation: reporting supported claims, communicating useful available information, and withholding conclusions when required evidence is missing. The question is how accurately the explanation tracks its evidence, rather than how successfully the vessel navigates. A correctly calibrated answer can diagnose an observed violation or state that completion is unknown; neither is a navigation-performance score.

We apply the same evidence-contract principle through marine-specific task contracts, odometry adapters and temporal computations. Land commands, motion and execution history govern supported diagnostic claims. Marine action status, adjacent pose/velocity measurements and a sampled five-second interval govern physical-completion claims. L0, L1 and L2 remove information from the same recording, testing whether explanation specificity grows with evidence while unsupported completion and causal claims remain withheld.

Existing evaluated answers provide direct evidence of explanation quality (Table I). Across five development recordings in three approach clusters, the marine calibrated implementation achieved 15/15 successful answers and 60/60 required information units. All 15 answers were assessed as supporting their inventoried claims and preserving causal limitations. In a separate explicit-contact replay over two existing recordings, all six answers passed development support/coverage criteria: 116/116 reviewed claim instances were supported, 24/24 common information units were communicated, and both answerable positive sampled-compliance intervals were explained. The banks remain separate; levels, claims and judge passes add no independent configuration N. The first bank uses project sentence inventories; its later incomplete atomic reassessment does not replace these scores.

The timeline/evidence-ladder figures show what those results mean. In the violation example (Fig. 1), the answer changes from unknown interval completion at L0 to an adjacent measurement at L1, then identifies a witnessed within-dwell yaw-rate violation at L2. It invents no physical cause. In the positive example (Fig. 2), L2 reports that all 251 observed kinematic/hull samples met the interval requirements, while retaining unknown contact and unproved continuous compliance. A later position crossing is outside that interval and cannot

TABLE I

Observed marine explanation quality in separate inspected development banks. Successful answers require supported inventoried claims, required-unit coverage and preserved limitations. These are automated support assessments, not population error bounds or a pooled domain-comparison estimate.

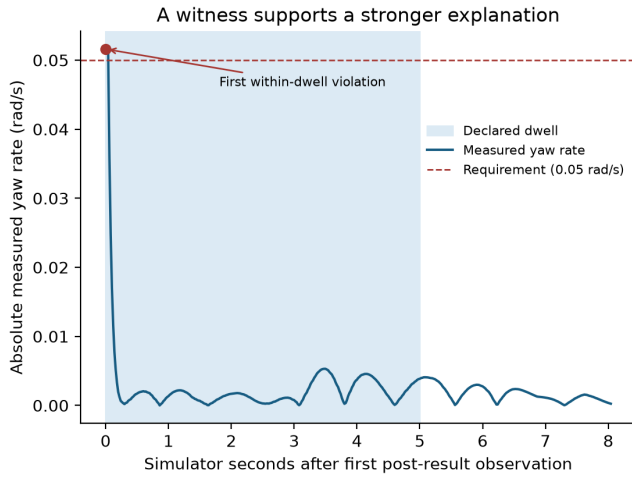
Evaluated bank	Answers	Required units
Five-record violation bank	15/15	60/60
Two-record contact replay	6/6	24/24

reverse its sampled-compliance statement. The excerpts are verbatim evaluated answers, rather than newly written ideal responses.

These results support transfer of the core calibration behavior to a practically motivated surface-vessel task, consistent with the land study’s supported-diagnosis, coverage and evidential-restraint objectives. They do not estimate equal land/marine accuracy or superiority over another method. The evaluated marine condition uses deterministic language and domain-specific computations, rather than an unchanged transfer of the full land pipeline. Labels are automated agent-assessed development evidence with possible correlated error, not human validation. Existing docking-policy explanations already use surrogate models [1]; the transferred behavior here concerns support for execution-completion claims. The demonstrated application is simulated; real-water accuracy and broader generalization remain to be tested.

References

- [1] V. B. Gjærum, I. Strümke, O. A. Alsos, and A. M. Lekkas, “Explaining a deep reinforcement learning docking agent using linear model trees with user adapted visualization,” *Journal of Marine Science and Engineering*, vol. 9, no. 11, p. 1178, 2021. [Online]. Available: <https://arxiv.org/abs/2203.00368>



Actual evaluated answer excerpts

L0: reported status is insufficient

"Physical docking completion is unestablished by the available evidence."

L1: accurate adjacent observation

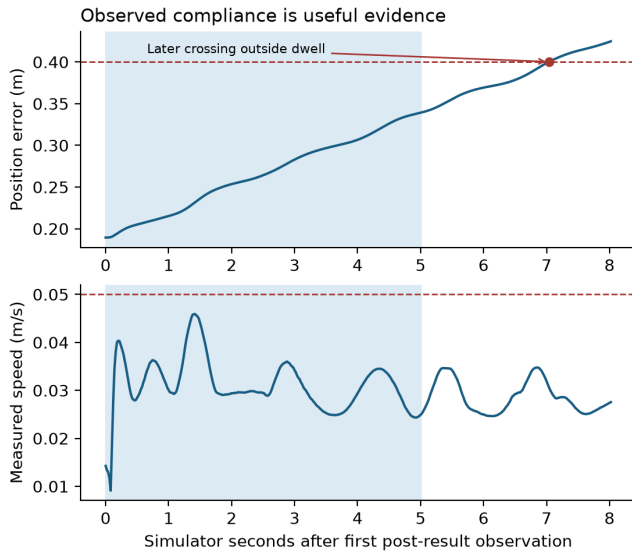
"The result-adjacent observation had position error 0.0945 m and measured speed 0.0175 m/s."

L2: supported violation diagnosis

"The specified docking dwell failed: at observed time 189.103 s, measured absolute yaw rate was 0.0516 rad/s, exceeding the 0.0500 rad/s bound."

Same physical record at L0-L2. The explanation diagnoses an observed requirement violation while leaving the physical cause unidentified.

Fig. 1. Evidence-sensitive diagnosis in the scored direct-route recording. The same-record answer moves from withholding interval completion to adjacent measurements and then a supported yaw-rate-violation witness. Text excerpts are verbatim from evaluated answers. Shading is the fixed dwell; time zero is its first post-result observation. The complete answers retain causal uncertainty.



Actual evaluated answer excerpts

L0: calibrated withholding

"Physical docking completion is unestablished by the available evidence."

L1: supported specificity

"The result-adjacent observation had position error 0.1893 m and measured speed 0.0146 m/s."

L2: useful positive information

"All 251 observed samples in the declared 248.420270-253.420270 s dwell met the position, heading, translational-speed, yaw-rate and hull-containment requirements (maximum sample gap 0.0200 s)."

Shading: fixed five-second dwell. Full completion stays unknown because contact is unavailable; samples do not prove continuous compliance.

Fig. 2. Useful positive evidence without an unsupported completion claim. The known-route contact-replay answer communicates sampled kinematic/hull compliance across the shaded fixed dwell. Its full text retains unknown contact and continuous-time limits. The later position crossing is outside the interval and does not falsify earlier sampled compliance. Excerpts are verbatim evaluated answers; plotted lines interpolate samples for display, not physical proof.

Marine explanation accuracy across removal-only evidence levels

L0: status only	L1: adjacent measurement	L2: sampled interval
Known route 15/15 reviewed claims supported 4/4 required units; limits preserved	Known route 18/18 reviewed claims supported 4/4 required units; limits preserved	Known route 25/25 reviewed claims supported 4/4 required units; limits preserved
Direct route 15/15 reviewed claims supported 4/4 required units; limits preserved	Direct route 18/18 reviewed claims supported 4/4 required units; limits preserved	Direct route 25/25 reviewed claims supported 4/4 required units; limits preserved

Six inspected answers; two existing recordings. Automated support review, not population accuracy or a comparison effect.

Fig. 3. All six calibrated answers in the closed explicit-contact replay. Each cell reports reviewed claim support and required-unit coverage for one recording/evidence level. The full answers preserve limitations. Claim counts are annotation denominators, not independent observations or an objective to maximize. The five-record bank and the replay are not pooled.